Surface Growth with Power-Law Noise

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We investigate a continuum formulation of surface growth with a power-law distribution of the magnitudes of regional advances. An exact theory predicts a sharp transition of the scaling behavior from power-law noise domination to a Gaussian noise regime as the power increases. Excellent agreement with simulations is obtained. This formulation describes Zhang's ballistic deposition model with power-law noise and possibly recent fluid displacement experiments.

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The description of the noise-driven growth of a self-affine interface far from equilibrium is a challenging problem. Self-affinity in this context means that the width $W(L,t)$ for an initially flat surface with lateral size L at time t can be expressed in the scaling form $W(L,t) = L^\alpha f(t/L^z)$, where α and z are called the roughness and dynamical exponents, respectively [1]. The scaling function f has asymptotic properties such that $W(L,t) \sim t^{\alpha/z}$ for $t \ll L^z$ and $W(L,t) \sim L^\alpha$ for $t \gg L^z$. This property occurs for many models of surface growth. An example is growth models in the universality class of the continuum equation of Kardar, Parisi, and Zhang (KPZ) [2], which should describe fluid flow, vapor deposition, and directed polymers. In 1+1 dimension, both theory and simulations give $\alpha = 0.5$ for bounded Gaussian noise. However, experiments on immiscible fluid displacement give α in the range 0.73–0.89 [3]. Zhang suggested an explanation by postulating that the noise, η, in the experiment follows a power-law distribution $\frac{1}{\eta^{\mu+1}}$. The exponents, α and z, are functions of μ [4]. The existence of this novel type of noise is supported by another recent fluid displacement experiment [5].

In addition to its possible experimental relevance, Zhang's model is important theoretically. For example, one should expect that α and z would be universal for any given μ for this model. This is not supported by numerical simulation and model-dependent values have been obtained [6, 7]. In fact, the existence of any continuum description has been put in doubt [8]. Further, the self-affinity of the interface itself has been questioned and multi-self-affinity suggested [9].

In this paper, we give an exact theory of growth with power-law noise which should clarify the situation. We show the surprising result that α is *exactly* given by the estimate of Zhang [8] and Krug [10]:

$$\alpha = \begin{cases} \alpha_p(\mu) & \text{for } d+1 \le \mu \le \mu_c , \\ \alpha_G & \text{for } \mu_c \le \mu , \end{cases} \tag{1}$$

in $d+1$ dimension, where $\alpha_p(\mu) = \frac{d+2}{\mu+1}$ and α_G is the roughness exponent for the KPZ equation with Gaussian noise. The critical μ_c is defined by $\alpha_p(\mu_c) = \alpha_G$. The dynamical exponent z is given by the identity [11, 12]

$$\alpha + z = 2 . \tag{2}$$

The apparent contradictions with previous numerical results [6, 7] have led to the general belief that Eq. (1) is only an estimate. We will show that Eq. (1) is exact and discuss how the apparent contradictions occur. More details will be reported elsewhere [13].

We first start with a simplified model which will be generalized subsequently. Consider the continuum model defined by

$$\frac{\partial h}{\partial t} = \frac{\lambda}{2}(\nabla h)^2 + \eta(x,t) , \tag{3}$$

where $h(x,t)$ is the interface height profile. The power-law noise $\eta(x,t)$ is a superposition of independent events:

$$\eta(x,t) = \sum_i g(\eta_i, x - x_i)\delta(t - t_i) , \tag{4}$$

where g corresponds to the advance of a whole region of the surface by η_i. We assume a scaling form

$$g(\eta, x) = \eta p((A^\sigma/\eta)^{1/\alpha} x) , \tag{5}$$

where A and σ are constants to be specified later and the noise profile p is any positive definite smooth function with bounded support maximized at $p(0) = 1$. The noise events have a uniform density $n(x,t,\eta)$ in space and time and a power-law distribution in magnitude, i.e.,

$$n(x,t,\eta) = \begin{cases} A\mu\frac{1}{\eta^{\mu+1}} & \text{for } \eta \ge \eta_m , \\ 0 & \text{for } \eta < \eta_m . \end{cases} \tag{6}$$

Therefore A controls the intensity of the noise and η_m is the lower cutoff.

Applying a scale transformation $x = bx'$, $t = b^z t'$, $h(x,t) = b^\alpha h'(x',t')$, $\eta(x,t) = b^\alpha \eta'(x',t')$, $\eta = b^\alpha \eta'$, and $\eta_m = b^\alpha \eta_m'$, the model defined by Eqs. (3)–(6) in the primed coordinates is invariant if we put $\alpha + z = 2$ [i.e., Eq. (2)] and $\alpha\mu = d + z$, which imply $\alpha = \alpha_p(\mu)$. We assumed that λ obtains no correction upon rescaling and will be justified. Using this scale invariance, the values of any given statistical measurement at different scales can be related. In particular, the steady-state surface width is now a function of L and η_m. We have

$$W(bL, b^{\alpha_p}\eta_m) = b^{\alpha_p} W(L, \eta_m) . \tag{7}$$

There exists a similar invariance property with respect to a change of the noise intensity A. Analogous calculations generalize Eq. (7) to

$$W \sim L^{\alpha_p} A^{\sigma_p} \tag{8}$$

provided we put

$$\eta_m \sim L^{\alpha_p} A^{\sigma_p} , \tag{9}$$

$$\sigma_p(\mu) = \frac{1}{\mu + 1} . \tag{10}$$

We stress that at the present stage where we scale the cutoff and the noise profile, the scaling properties are *not* approximations or asymptotic results but are *exact* for all L and A and for all $\mu \geq d+1$, even beyond μ_c.

We now add more terms to Eq. (3) to produce a generalized KPZ equation with power-law regional advance:

$$\frac{\partial h}{\partial t} = \nu \nabla^2 h + \frac{\lambda}{2} (\nabla h)^2 + \xi(x,t) + \eta(x,t) , \tag{11}$$

where $\xi(x,t)$ is the uncorrelated Gaussian noise. Suppose that $\mu < \mu_c$, i.e., $\alpha_p > \alpha_G$, and that we scale the system according to the usual KPZ exponents α_G and z_G; it is easy to see that the power-law noise intensity A increases under coarsening. Therefore at sufficiently large length scales, the surface is effectively "stretched" compared to the pure KPZ case. This stretching upsets the balance of the terms which have the same scaling behavior in the ordinary KPZ case. That is, by simple power counting, $\nu \nabla^2 h$ and ξ are irrelevant compared to $\frac{\lambda}{2} (\nabla h)^2$. This can be explicitly demonstrated in 1+1 dimension using the renormalization flow [14] equations in Ref. [2]. They are directly applicable here because the lower cutoff is scaled in the model itself so that the high-frequency modes of $\eta(x,t)$ need not be integrated out and moreover the smooth noise profile being in the long-wavelength limit is not renormalized. The coefficient λ is also not renormalized due to the invariance of Eq. (11) under an infinitesimal tilting similar to the original KPZ equation [12]. As a result, Eq. (11) reduces to Eq. (3) and falls in the same universality class. At any finite scale, this causes a crossover. If A is small, so that the power-law noise is negligible, we expect KPZ behavior. However, upon coarse graining, so that Eq. (3) is approached, power-law noise dominates and Eq. (8) applies. The effective roughness exponent $\alpha(L)$ for system size L thus crosses over from α_G to α_p as L increases. In a similar way, for $\mu > \mu_c$, A decreases under rescaling so $\alpha(L)$ crosses over from α_p to α_G instead. This proves Eq. (1) for this case.

We verify the above results numerically for 1+1 dimension using a ballistic deposition model with power-law regional surface advances. We take the basic interface evolution rule with Gaussian noise to be

$$h(x,t+1) = \tfrac{1}{2} [\max\{h(x,t), h(x+1,t), h(x-1,t)\} + h(x,t)] + \xi(x,t) , \tag{12}$$

where $\xi(x,t)$ is a uniform random deviate in the range 0 to 1. Motivation for this unusual choice will be given below. Periodic boundary conditions and a parallel updating algorithm are used so that growth occurs at all even (odd) sites at even (odd) time steps. Power-law noise occurs independently with much lower probability according to Eq. (6). If power-law noise η_i occurs at a site x_i, a whole neighborhood advances instantly from h to $h + g(\eta_i, x - x_i)$. We take a truncated parabolic profile function $p(y) = \max\{1 - [y/(4 \times 0.015^{1/3})]^2, 0\}$ and $\eta_m = (\frac{L}{16})^{\alpha_p} (\frac{A}{0.015})^{\sigma_p}$.

Figure 1 shows a log-log plot of W against L for $\mu = 2$ to 6 for a subset of our data with $A = 0.015$. Also shown is the pure Gaussian noise case ($A = 0$) which bounds the other curves from below. The local slopes of the curves correspond to $\alpha(L)$. We can see the two different crossovers mentioned above separated by the critical $\mu_c = 5$ case. To check Eq. (8), we have to adjust A to go beyond the crossover region to the power-law noise dominated regime. To do this, we need W to be significantly larger than for $A = 0$. However, if W is too large, examination of the interface reveals that microscopic roughness is smoothed out due to the strong stretching. The interface is in a model-dependent "anomalous growth mode" which is incompatible with the continuum description [15] and resembles step flow in this case. Therefore Eq. (8) is approached only for a narrow window; see Fig. 1. At large L and intermediate values of A corresponding to this region, the fitted exponent α is shown in Fig. 2. $\alpha = \alpha_p(\mu)$ is nicely verified for $\mu \gtrsim 3$. The discrepancy for $\mu \simeq 2$ is due to the remnant of the crossover which is more pronounced for $\alpha_p(\mu)$ very different from α_G. The transition stated in Eq. (1) is not evident here because we are showing results in the power-law noise dominated regime instead of the asymptotic regime.

The whole data set is presented in Fig. 3, where $W/L^{\alpha_p(\mu)}$ is plotted against A on a log-log scale. The arrows point to the subset of data for Fig. 2, where the

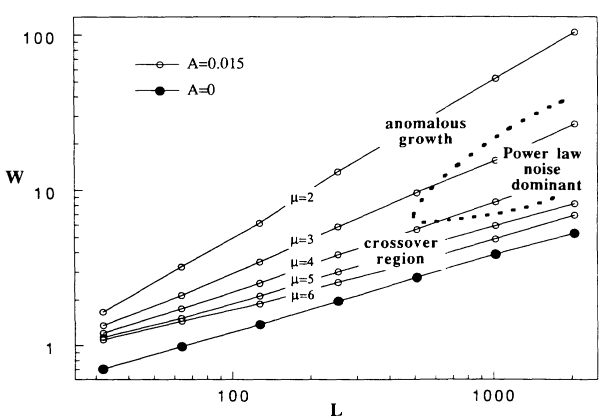

FIG. 1. Saturated surface width W as a function of system size L for $\mu = 2$ to 6 in order of decreasing W.

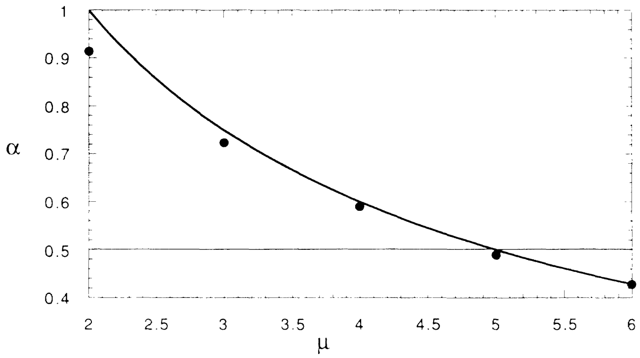

FIG. 2. Roughness exponent α as a function of power of the noise distribution μ in power-law noise dominated regime. The solid line is the theoretical curve $\alpha_p(\mu)$.

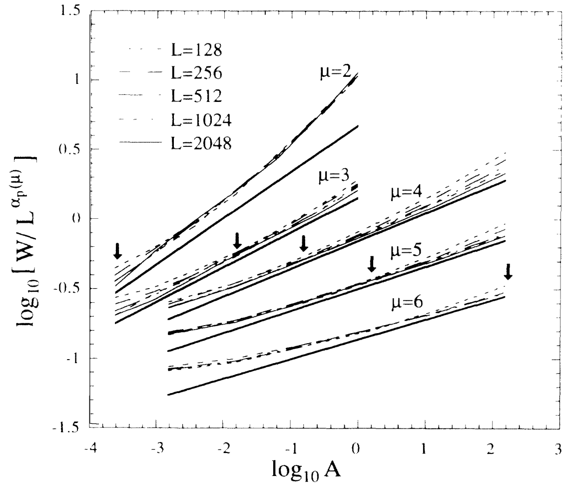

FIG. 3. Rescaled surface width $W/L^{\alpha_p(\mu)}$ against noise intensity A on a log-log scale for $\mu = 2$ to 6. The vertical scale is shifted by 1.5, 0.8, 0.4, 0, and -0.4, respectively. The arrows and the solid lines are explained in the text.

good data collapse for $\mu \gtrsim 3$ corresponds to the fact that $\alpha = \alpha_p(\mu)$. The good agreement with the expected slope (thick solid lines) verifies Eq. (10). For $\mu = 3$ and 4, both the data collapse and the expected slope break down for small A due to crossover and for large A due to anomalous growth, while they hold for a narrow range of A which widens as L increases. For the critical $\mu = 5$, data collapse is good for intermediate values of A down to 0 in spite of the crossover because $\alpha_p(5) = \alpha_G$. For $\mu = 2$, the collapse is good for large A because it happens accidentally that anomalous growth also gives $\alpha = 1$ [16]. However, the expected slope σ_p does not hold, showing that the continuum description is not valid. Previous simulations [6, 7] are in this regime so that the discrepancy shown in Fig. 2 for $\mu \simeq 2$ is not observed.

We have so far proved formulas (1), (2), and (10) for the generalized KPZ equation (11). We will now further generalize the results. In Zhang's model, the evolution is defined by an equation similar to Eq. (12) with $\xi(x,t)$ replaced by the power-law variable η. We seek to describe it using Eq. (11) by defining an effective lower cutoff η_{m0}. Large values of the noise, with $\eta > \eta_{m0}$, correspond to the rare large fluctuations represented by the $\eta(x,t)$ term in Eq. (11). The truncated lower part of the spectrum with $\eta < \eta_{m0}$ behaves like Gaussian noise and can be represented by the $\xi(x,t)$ term.

We are thus led to consider a constant cutoff, $\eta_m = \eta_{m0}$, instead of Eq. (9). For $\mu < \mu_c$, assume that L is large enough so that Eq. (11) reduces to Eq. (3) and hence Eq. (7) holds. Rearranging Eq. (7) gives

$$W(L,\eta_{m0}) = (L/L_0)^{\alpha_p} W(L_0, \eta_m(L)) \,, \tag{13}$$

where $\eta_m(L) = (L_0/L)^{\alpha}\eta_{m0}$. As $L \to \infty$, $\eta_m(L) \to 0$. The term $\lim_{\eta_m \to 0} W(L_0,\eta_m)$ approaches a constant so that Eq. (8) still holds asymptotically. This is because if the above limit were to explode, W could be made arbitrarily large by decreasing η_m and the roughness would be dominated by the part of the power-law noise spectrum arbitrarily close to zero. This implies that the large fluctuations are irrelevant which is not true in this case. For $\mu > \mu_c$, the large fluctuations are irrelevant compared to both the Gaussian noise and the smaller power-law noise and hence $\alpha = \alpha_G$. Therefore, all our scaling formulas remain intact. At the critical $\mu = \mu_c$, we postulate that the limit diverges logarithmically so that Eq. (13) gives

$$W(L,\eta_{m0})^2 \sim L^{2\alpha_G} \ln(L) \,. \tag{14}$$

This logarithmic correction is motivated by a similar result for a linear model [13] and is indeed the principal reason why criticality was previously overlooked [4, 6]. We simulated Zhang's model using Eq. (12) for $\mu = 5$. A fit of the measured surface width to the form $W(L) \sim L^{\alpha}$ for $L = 128$ to 4096 gives $\alpha \simeq 0.59$, consistent with previous results [6]. However, fitting over ranges from $\frac{L}{2}$ to $2L$ gives the L-dependent effective $\alpha(L)$ shown in Fig. 4, implying that $\alpha = \lim_{L\to\infty} \alpha(L) \ll 0.59$. The $W(L)$ curve is in fact well described by Eq. (14) as demonstrated in the inset supporting $\alpha = 0.5$.

The result $\alpha \simeq 0.5$ for $\mu = 5$ has been obtained numerically using sequential updating [7]. Our simulations show that this introduces another large correction due to the larger "intrinsic width." This correction might partially cancel the one due to the cutoff but significantly increases the uncertainty in the measured α.

Finally we generalize the result to other noise profiles. Consider replacing Eq. (5) by

$$g(\eta, x) = \eta p(\eta^{-1/\gamma} x) \,, \tag{15}$$

where $\gamma > \alpha$. Upon the previous scale transformation, Eq. (15) is no longer invariant as is Eq. (5) but $p(y)$ has to be replaced by $p(b^{1-\alpha/\gamma} y)$. As $b \to \infty$, $p(y)$ approaches the fixed point $p^*(y) = \{1 \text{ if } y = 0;\ 0 \text{ if } y \neq 0\}$. Invariance is then restored and our scaling formulas are applicable again. Similar arguments can be applied to the

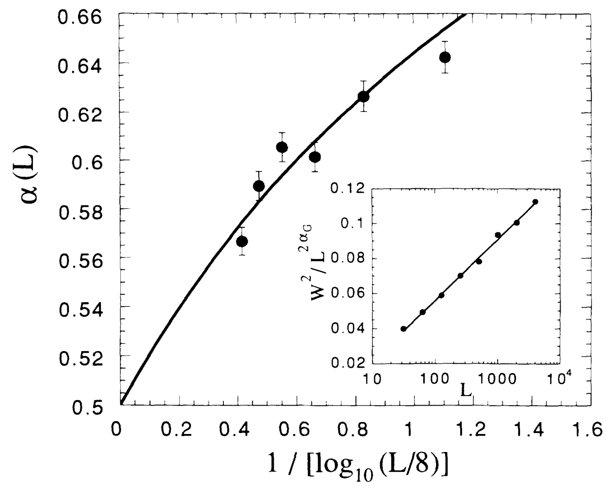

FIG. 4. Effective roughness exponent $\alpha(L)$ against system size L for $\mu = 5$. The solid line is derived from the linear fit in the inset. Inset: Correction to scaling $W^2/L^{2\alpha_G}$ against L on a log-linear scale.

general case where g is any given stochastic single peaked positive smooth function satisfying $g(\eta, 0) \sim \eta$ with the widths of the peaks bounded by $\eta^{1/\alpha}$ as $\eta \to \infty$.

For Zhang's model, an initial large fluctuation η creates huge roughness on the surface, leading to anomalous growth. Only after the fluctuation diffuses so that the local slope everywhere is restored to moderate values will the continuum description be valid again. The advance of the interface during this anomalous growth period less the average growth can be set to $g(\eta, x)$. By examining growth rules like Eq. (12) and similar variants, we see that g defined in this way can easily satisfy the criteria specified above. In addition, the anomalous growth period is negligible compared to the dynamical time scale of the interface for large scales, justifying the assumption of instantaneous regional surface advances. Therefore our theory should apply also for Zhang's model. Because of the very loose restriction of its applicability, it is possible that this theory also describes the fluid displacement experiments provided there exists, for some microscopic reason, a power-law noise.

At finite scales, the anomalous growth period is not negligible. This can cause another important crossover. Equation (12) and the very smooth parabolic noise profiles are chosen because they minimize the anomalous growth as observed by examining the aggregate. Using other ballistic deposition rules or profiles can lead to slightly different values in the measurement of α together with noticeable anomalous growth. However, the qualitative features are the same as reported here. Most growth rules might also lead to the previously reported multi-self-affinity [9] during the crossover which can be suppressed simply by adopting Eq. (12) [13].

Our approach has been first to consider power-law noise with an explicitly scaled lower cutoff so that its renormalization is avoided. One may be tempted to apply the standard perturbative renormalization-group technique [2] directly to the power-law noise. In this method, when we approach the fixed point, the power-law noise will attain the profile $p^*(y)$ (see above) with a finite lower cutoff in magnitude. This implies that it has measure zero and disappears readily under the $\nu\nabla^2 h$ term. However, it should be able to give finite fluctuations because $\nu = 0$ at the fixed point when the power-law noise is relevant. As a result, a perturbative scheme with $\nu\nabla^2 h$ being the unperturbed term might not accommodate power-law noise in the current context.

In conclusion, we give an exact scaling theory for a generalization of the KPZ equation with power-law noise which should describe the asymptotic behavior of Zhang's power-law noise model and possibly fluid displacement experiments, though the justification of the existence of any power-law noise in the experiment is beyond the scope of this work. The apparent contradiction of our theory with previous simulations are due to crossover effects.

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